A Comparative Study of Transformer and Hybrid Models for Fake News Detection

A thesis submitted in partial fulfilment of the requirements for the degree of

Master of Computer Application

by

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Dedicated to my parents

Abstract

This thesis investigates automated fake news detection using Natural Language Processing (NLP) and Machine Learning (ML). It compares Bidirectional Long Short-Term Memory (BiLSTM), fine-tuned Bidirectional Encoder Representations from Transformers (BERT), and a hybrid BERT + BiLSTM model on benchmark datasets, including LIAR and ISOT. The framework combines text preprocessing, representation, and classification; the hybrid feeds BERT's contextual token embeddings into a BiLSTM layer.

Under the reported training conditions, BiLSTM achieves 50% accuracy and an F1-score of 0.3333, while BERT achieves 95% accuracy, precision of 0.9545, and recall of 0.9500. The hybrid matches BERT without measurable improvement. The study attributes BiLSTM's limited performance to few training epochs and restricted hyperparameter tuning. These findings support transformer-based representations under the conditions examined.

Future extensions include Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Explainable AI (XAI) for evidence grounding and interpretation.

Keywords: Fake News Detection, Natural Language Processing, Machine Learning, Deep Learning, BERT, BiLSTM, Transformers, Explainable AI, Misinformation.

Declaration

I certify that

- The work contained in the dissertation is original and has been done by myself under the general supervision of my supervisor.
- The work has not been submitted to any other Institute for any degree or diploma.
- I have followed the guidelines provided by Tripura University in writing the thesis.
- I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the university.
- Whenever I have used materials (data, theoretical analysis, and text) from other sources, I have given due credit to them by citing them in the text of the dissertation and giving their details in the references.
- I have not use any AI websites (eg: chat GPT) for paraphrasing sentences.
- I have not plagiarised any content from other sources.

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All helps received by him from various sources have been duly acknowledged. No part of this thesis has been submitted elsewhere for award of any other degree.

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Acknowledgment

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Chapter 1

Introduction

Fake news detection is the automated process of identifying and categorizing false, misleading, or fabricated information that is deliberately or unintentionally portrayed as genuine news, using computational methods drawn from Natural Language Processing (NLP) and Machine Learning (ML) [14, 15]. The emergence and rapid growth of digital media platforms has led to an unprecedented increase in the volume, velocity, and reach of information generated and shared online every day. News, opinion, and rumour now travel through the same channels, often indistinguishable from one another at first glance, and readers are left to judge credibility with little more than a headline and a few seconds of attention.

This state of affairs creates serious challenges, including the manipulation of public perception, interference with democratic processes such as elections, and a gradual erosion of confidence in previously trusted sources of information [14, 15]. Empirical studies have shown that false information spreads significantly faster and reaches a wider audience on social media than verified, factual news [17]. Given this pace, it becomes practically infeasible for human journalists and professional fact-checkers to manually verify every claim before it spreads beyond control.

The consequences of unchecked misinformation extend well past individual misunderstanding; large-scale disinformation campaigns have been linked to social unrest, financial market manipulation, and public health harms. Consequently, building automated systems that can detect fake news reliably, at scale, and in near real time has become an important and urgent research problem, motivating the work undertaken in this thesis.

1.1 Challenges and Issues

The term "fake news" is not a single, homogeneous category of content. It is an umbrella term that spans multiple distinct classes of information, each with its own motive, structure, and linguistic signature. Broadly, the literature identifies five recurring types of fake news:

- **False News Stories:** Entirely fabricated information deliberately designed to deceive readers.
- **Manipulated News:** Genuine information that has been selectively edited, distorted, or taken out of context.
- **Satirical News:** Exaggerated or humorous content that is not intended to deceive but may be mistaken for factual reporting.
- **Clickbait:** News stories with sensational or misleading headlines that do not match their actual content.
- **Propaganda:** Biased or slanted information deliberately produced to advance a particular agenda or cause.

This diversity is precisely what makes automatic detection difficult. A model trained to recognize the linguistic markers of one category, such as sensational clickbait headlines, may fail entirely when confronted with subtly manipulated but factually-grounded reporting, or with satire that mimics a legitimate news outlet's tone. Detecting fake news therefore requires models that can capture deep contextual and semantic relationships in text, rather than relying on superficial keyword or stylistic cues alone.

A second layer of difficulty comes from the modelling side. Classical machine learning algorithms such as Logistic Regression, Support Vector Machines (SVM), Naive Bayes, and Random Forest have historically been applied to this task using hand-engineered statistical features (e.g., TF-IDF vectors). These models are simple, interpretable, and computationally cheap, but they cannot capture word order or long-range semantic dependency, which limits their ceiling on accuracy. Sequential deep learning models such as Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM) networks were introduced to address this gap by explicitly modelling sequences of words, yet their effectiveness can be undermined when contextual relationships are complex, when training data is limited,

or when training time and hyperparameter tuning are restricted – a limitation observed under the restricted training conditions used in this study.

Transformer-based models, most notably BERT (Bidirectional Encoder Representations from Transformers), have been shown to outperform these earlier sequential approaches by producing deeply bidirectional, contextualized representations of text in a single forward pass [4, 7]. Where sequential models process text step by step, BERT attends to the entire input context simultaneously through self-attention, giving it a considerably richer view of semantic relationships between words. This contrast between sequential and attention-based contextual modelling is one of the central technical challenges this thesis investigates.

1.2 Motivation of the Research

The motivation for this research is twofold. First, from a societal standpoint, the scale and speed at which misinformation now spreads make manual fact-checking an inadequate defence on its own; an automated, scalable first line of detection is necessary to support (not replace) human fact-checkers and platform moderators. Second, from a technical standpoint, the field has moved rapidly from classical statistical text representations (Bag-of-Words, TF-IDF) through static embeddings (Word2Vec, GloVe) to deep contextual embeddings (BERT), and it is not always clear, for a given dataset and task, whether the additional complexity of transformer and hybrid architectures is actually justified by a corresponding improvement in accuracy.

This thesis is therefore motivated by the need to systematically and empirically compare a purely sequential deep learning model (BiLSTM), a purely transformer-based model (BERT), and a hybrid model that combines both (BERT + BiLSTM), across multiple benchmark datasets, so that the trade-offs between model complexity, training cost, and detection performance can be clearly understood. Such a comparison directly informs practical deployment decisions for real-world fake news detection systems, including social media monitoring pipelines and automated fact-checking assistants.

1.3 Research Objectives

The specific objectives of this research are as follows:

1. To design a multi-stage fake news detection pipeline that integrates NLP-based text preprocessing, multiple feature extraction/representation techniques, and a classification layer.

2. To implement and train three classification models – a standalone BiLSTM network, a fine-tuned BERT model, and a hybrid BERT + BiLSTM model – under comparable training conditions.

3. To evaluate and compare the three models using standard classification metrics (accuracy, precision, recall, and F1-score) as well as visual diagnostics (confusion matrices, ROC curves, precision–recall curves, and loss curves).

4. To assess the generalization ability of the proposed models across multiple benchmark datasets, namely LIAR and ISOT.

5. To identify the practical and architectural trade-offs between sequential, transformer-based, and hybrid models for fake news detection, and to outline directions for future improvement using large language models and explainable AI.

1.4 Thesis Contributions

This thesis makes the following key contributions:

- Proposes a layered architecture for fake news detection that combines NLP preprocessing, multiple feature extraction techniques, and a comparative classification framework.

- Presents a systematic experimental comparison of BiLSTM, BERT, and a hybrid BERT + BiLSTM model across multiple benchmark datasets.

- Demonstrates, with quantitative evidence, the superiority of transformer-based models over purely sequential deep learning models for this task under constrained training budgets.

- Investigates and documents the specific limitations of sequential models and their hybrid variants when trained with a small number of epochs and without extensive hyperparameter tuning.

- Identifies promising directions for future research, including the integration of Large Language Models (LLMs), Retrieval-Augmented Generation

(RAG), and Explainable AI (XAI) techniques into fake news detection pipelines.

1.5 Thesis Organization

This thesis is organized into three chapters.

Chapter 1: Introduction

This chapter introduces fake news detection, the challenges and motivation for the research, and the objectives and contributions of the thesis.

Chapter 2: Related Work

This chapter reviews previous research on classical machine learning, deep learning, hybrid models, benchmark datasets, large language models, and explainability.

Chapter 3: Proposed Approach

This chapter describes the dataset, data preprocessing, system pipeline, and construction of the BiLSTM, BERT, and hybrid BERT + BiLSTM models.

Chapter 2

Related Work

2.1 Introduction

This chapter reviews nineteen peer-reviewed articles that form the basis of this research. The selected articles have been organized thematically, starting with traditional machine learning techniques, moving through deep learning and hybrid architectures, evidence-aware and attention-based methods, benchmark dataset and survey papers, and finally the role of large language models and explainability. The following discussion groups these articles by their methods and research focus.

Early work on automated fake news detection relied heavily on classical machine learning classifiers trained over statistical text features. Aphiwongsophon and Chongstitvatana [1] systematically compared Logistic Regression, SVM, Naive Bayes, and Random Forest using TF-IDF features, concluding that such approaches are simple and computationally cheap but consistently trail deep learning methods in raw accuracy because they cannot capture word order or contextual meaning. Huang and Chen [2] pushed this line of work further by using a meta-heuristic harmony search algorithm for automatic feature selection and ensemble weight optimization, improving robustness at the cost of significant computational overhead. Przybyla [10] took a complementary stylometric approach, showing that writing-style features such as sentence length and punctuation patterns can distinguish fake from credible news without any external knowledge base, though such style-based signals are also the easiest for a sophisticated adversary to mimic.

A second wave of research moved from statistical features to learned representations using deep neural networks. Nasir et al. [8] proposed a hybrid CNN-

RNN architecture with GloVe embeddings, arguing that combining local (CNN) and sequential (RNN) feature extraction improves generalization over single-model approaches, though at the cost of slower inference and reduced effectiveness on very short texts such as tweets. Ruchansky et al. [13] extended this idea beyond pure text, proposing the CSI (Capture, Score, Integrate) model, which was among the first to jointly model article content, user response behaviour, and propagation patterns – a reminder that text-only models such as the ones evaluated in this thesis omit potentially useful social-context signals. Popat et al. [9] introduced an evidence-aware detection approach that retrieves external web evidence and combines it with an attention-based LSTM for credibility scoring, improving interpretability but introducing a dependency on the availability and reliability of retrieved evidence.

Closest in spirit to the model proposed in this thesis is the work of Rai et al. [11], who combined BERT contextual embeddings with an enhanced LSTM architecture and an additional attention mechanism, reporting performance gains over standalone BERT or LSTM models on the LIAR and ISOT datasets. Kaliyar et al. [4] similarly combined BERT with parallel CNN layers (FakeBERT), reporting up to 98.90% accuracy, at the cost of substantial GPU memory and compute requirements. These works motivate the hybrid BERT + BiLSTM architecture investigated in Chapter 3 of this thesis, while also cautioning that hybridization does not automatically guarantee an improvement over a well-tuned standalone transformer..

Several works in the literature focus not on new models but on datasets, taxonomies, and empirical characterization of the misinformation problem itself. Wang [18] introduced the LIAR benchmark dataset, comprising 12,836 manually labelled short political statements from PolitiFact with fine-grained truthfulness labels and rich metadata; LIAR is one of the two benchmark datasets used for evaluation in this thesis. Shu et al. [15] released FakeNewsNet, a multi-modal dataset combining news content, social engagement, and propagation features across PolitiFact and GossipCop, while Shu et al. [14] and Zhou and Zafarani [19] each provide broad taxonomies of detection approaches spanning knowledge-based, style-based, propagation-based, and credibility-based methods. Mehta et al. [7] surveys more recent deep-learning-era techniques, including transformers, graph neural networks, and multi-modal approaches, identifying gaps that this thesis partially addresses through its systematic sequential-vs-transformer-vs-hybrid comparison. Separately, Vosoughi et al. [17] conducted a landmark large-scale empirical study of true and false news diffusion on Twitter, finding that false news spreads roughly

six times faster than true news – a finding frequently cited as core motivation for automated, real-time fake news detection research, including this thesis.

A final cluster of reviewed work looks beyond the BERT-era architectures studied experimentally in this thesis, toward large language models (LLMs) and explainability techniques that are identified as future research directions for extending this work. Touvron et al. [16] introduced LLaMA 2, an open-source family of LLMs that could be fine-tuned on fake news data or combined with retrieval pipelines. Lewis et al. [5] proposed Retrieval-Augmented Generation (RAG), which grounds LLM outputs in retrieved evidence from external knowledge bases, directly applicable to grounding fake news classification decisions in verified fact-checking databases such as Snopes or PolitiFact. On the interpretability side, Ribeiro et al. [12] and Lundberg and Lee [6] introduced LIME and SHAP respectively – model-agnostic explanation techniques that could be layered on top of a BERT-based fake news classifier to highlight which words or phrases drove a given REAL/FAKE prediction, directly addressing the black-box nature of the deep learning models evaluated in this thesis.

2.2 Summary

Taken together, this body of literature shows a clear progression from statistical, feature-engineered classical ML models, through hybrid deep learning architectures combining CNNs, RNNs, and external evidence, to transformer-based and BERT-hybrid models that dominate current state-of-the-art results, alongside a growing interest in large language models and explainability. It also surfaces two recurring open gaps that this thesis directly targets: (i) a lack of controlled, apples-to-apples comparison between purely sequential (BiLSTM), purely transformer-based (BERT), and hybrid (BERT + BiLSTM) models under identical training conditions and across multiple benchmark datasets, and (ii) limited attention to whether the added complexity of hybrid architectures is empirically justified. Chapter 3 describes the proposed system that is designed to address these gaps.

Chapter 3

Proposed Approach

3.1 Introduction

This chapter describes the fake news detection framework, its datasets and preprocessing steps, and the construction of the BiLSTM, BERT, and hybrid BERT + BiLSTM models.

The proposed detection system is designed as a multi-stage pipeline that converts unstructured news input into a classified output with an associated confidence estimate. Figure 3-1 illustrates the overall system architecture.

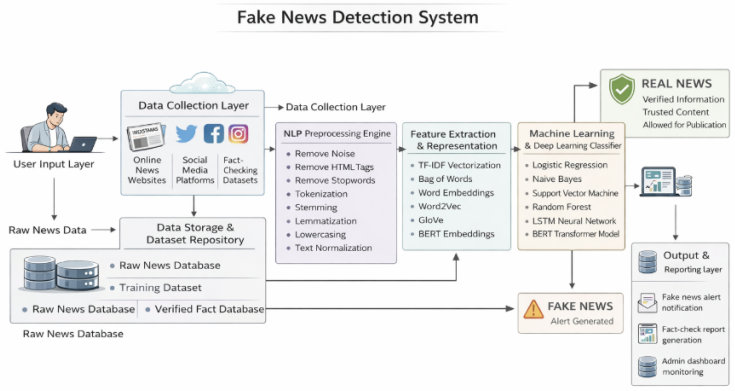


Figure 3-1: End-to-end fake news detection system pipeline, from user input through data collection, NLP preprocessing, feature extraction, classification, and output reporting

The pipeline is organized into the following stages:

- **User Input Layer:** Accepts raw news input in the form of free text, a headline, or a URL pointing to a news article.
- **Data Collection Layer:** Gathers the corresponding data from benchmark datasets (LIAR, ISOT, FakeNewsNet) and, where applicable, online news websites, social media platforms, and fact-checking sources.
- **Data Storage and Dataset Repository:** Persists raw news data, training/validation/test splits, and verified fact databases for reuse across experiments.
- **NLP Preprocessing Engine:** Cleans and normalizes the raw text through noise removal, HTML-tag stripping, stopword removal, tokenization, stemming, lemmatization, and lowercasing, as described in Section 3.3.
- **Feature Extraction and Representation Layer:** Converts the normalized text into numeric representations using TF-IDF, Bag-of-Words, Word2Vec, GloVe, and/or BERT embeddings, depending on the downstream classifier.
- **Machine Learning and Deep Learning Classifier:** Hosts the candidate classification models compared in this thesis – BiLSTM, BERT, and the hybrid BERT + BiLSTM model – and produces a REAL/FAKE prediction with a confidence score.
- **Output and Reporting Layer:** Emits the final classification decision together with a human-readable report, suitable for downstream use cases such as a fake-news alert notification, a fact-check report, or an administrative monitoring dashboard.

This layered design cleanly separates preprocessing, representation, and modelling concerns, which makes it straightforward to swap in the three candidate classifiers (BiLSTM, BERT, BERT + BiLSTM) without changing the surrounding pipeline.

3.2 Dataset Description

All experimental evaluations are performed using benchmark datasets widely used in the fake news detection literature: LIAR [18], ISOT [3], and FakeNewsNet [15].

Table 3.1 summarizes the size and composition of each dataset. A Kaggle-hosted version of these benchmark datasets was used for training and evaluation.

Table 3.1: Dataset Description

Dataset	Total Samples	Real News	Fake News	Source
LIAR	12,836	6,418	6,418	PolitiFact Statements
ISOT	44,919	21,417	23,502	Reuters & Fake News Sites
FakeNewsNet	23,196	11,941	11,255	PolitiFact & GossipCop

Each dataset was split in an 80:10:10 ratio, corresponding to the training, validation, and testing sets respectively, in accordance with established conventions in the literature [4, 11].

3.3 Data Preprocessing

Raw news text is noisy: it contains inconsistent casing, punctuation, HTML artifacts, and words that carry little discriminative signal. Before any feature extraction or model training can take place, the text must be cleaned and normalized. The standard NLP preprocessing pipeline used in this thesis consists of the following steps:

- **Noise Removal:** Stripping HTML tags, special characters, URLs, and other non-textual artifacts from the raw input.
- **Lowercasing:** Converting all text to a consistent case so that, e.g., "News" and "news" are treated identically.
- **Tokenization:** Splitting text into individual tokens (words or sub-word units), which form the atomic units consumed by downstream models.
- **Stopword Removal:** Removing high-frequency, low-information words (e.g., "the", "is", "and") that do not contribute to distinguishing fake from real news.
- **Stemming and Lemmatization:** Reducing words to their root or base form (e.g., "running" $\rightarrow$ "run"), which reduces vocabulary size and helps models generalize across word forms.
- **Padding / Truncation:** Normalizing the length of tokenized sequences to a fixed maximum length so that they can be processed in fixed-size batches by neural network models.

For transformer models such as BERT, tokenization is performed using a model-specific sub-word tokenizer (WordPiece), and two special tokens are introduced: a classification token, `[CLS]`, prepended to every sequence and used to aggregate sequence-level information, and a separator token, `[SEP]`, used to mark sequence boundaries. Each tokenized sequence is accompanied by an attention mask that indicates which positions correspond to real tokens (1) versus padding (0), so that the model does not attend to padded positions.

3.4 Model Building

All three models were trained using hyperparameters appropriate to their architecture, so that the comparison remains fair while respecting each model family's characteristics. BERT-based models use a lower learning rate and smaller batch size than BiLSTM, since they are already pre-trained and only require fine-tuning, whereas the standalone BiLSTM model, trained from scratch, uses a higher learning rate. Table 3.2 lists the exact configuration used for each model.

Table 3.2: Hyperparameter Configuration of Models

Hyperparameter	**BiLSTM**	**BERT**	**BERT + BiLSTM**
Embedding Type	GloVe	BERT Base	BERT Base
Max Sequence Length	200	256	256
Hidden Units	128	—	128
Number of Layers	2	12 (Transformer)	12 + BiLSTM
Dropout Rate	0.5	0.1	0.3
Batch Size	32	16	16
Learning Rate	0.001	2e-5	2e-5
Optimizer	Adam	AdamW	AdamW
Epochs	3	3	3
Loss Function	Binary Crossentropy	Crossentropy	Crossentropy

3.4.1 Fake News Detection Using BiLSTM

Long Short-Term Memory (LSTM) networks are a class of Recurrent Neural Networks (RNNs) specifically designed to model sequential data while mitigating the vanishing gradient problem that plagues vanilla RNNs over long sequences. An LSTM cell maintains an internal memory state that is selectively updated, forgotten, and output at each time step through a set of learned gates (input, forget,

and output gates), allowing the network to retain relevant information over longer spans of text than a standard RNN.

A standard LSTM processes a sequence in a single direction, typically left to right, meaning that at any given position, the model has only seen the preceding context. A Bidirectional LSTM (BiLSTM) addresses this limitation by running two LSTM layers over the input sequence in parallel, one processing it left-to-right and the other right-to-left, and concatenating (or otherwise combining) their hidden states at each position. This gives the BiLSTM access to both past and future context, which is particularly useful for understanding a word's role in a sentence.

Despite this improvement, BiLSTM networks still process text strictly sequentially at inference time, which limits parallelization during training, and their capacity to model very long-range dependencies remains weaker than attention-based architectures. Furthermore, BiLSTM models trained with a limited number of epochs and without extensive hyperparameter tuning can converge poorly, resulting in substantially lower accuracy than transformer-based alternatives.

3.4.2 Fake News Detection Using BERT

The Transformer architecture dispenses with recurrence entirely and instead relies on a self-attention mechanism, which allows every token in a sequence to directly attend to every other token in a single layer, regardless of their distance in the sequence. This removes the sequential bottleneck of RNN-based models and enables far greater parallelization during training, while also making it easier to model long-range dependencies.

BERT (Bidirectional Encoder Representations from Transformers) is built from a stack of Transformer encoder layers (12 layers in the BERT-Base configuration) and is pre-trained on large unlabelled text corpora using two self-supervised objectives: masked language modelling (predicting randomly masked tokens from their context) and next sentence prediction. This pre-training allows BERT to learn deep, bidirectional, general-purpose language representations before it is ever exposed to a downstream task.

For a downstream classification task such as fake news detection, a pre-trained BERT model is *fine-tuned*: the input news text is tokenized and fed through the pre-trained encoder, producing a contextual embedding for every

token, including the special [CLS] token, which is conventionally used as an aggregate representation of the entire sequence. A lightweight classification head (typically a dropout layer followed by a fully connected layer with a softmax or sigmoid activation) is then trained on top of this representation to predict the real/fake label, while the underlying BERT weights are also updated (fine-tuned) using a comparatively small learning rate, since they are already well-initialized from pre-training.

In the hybrid architecture examined in this thesis (Section 3.4.3), rather than relying solely on the [CLS] token, the entire sequence of contextual token embeddings produced by BERT is passed into a BiLSTM layer, allowing the network to additionally model sequential dependencies over BERT's already contextualized representations before final classification.

3.4.3 Fake News Detection Using BERT + BiLSTM

While the system model in Figure 3-1 shows the pipeline at a high level, Figure 3-2 zooms into the detailed architecture of the hybrid BERT + BiLSTM model.

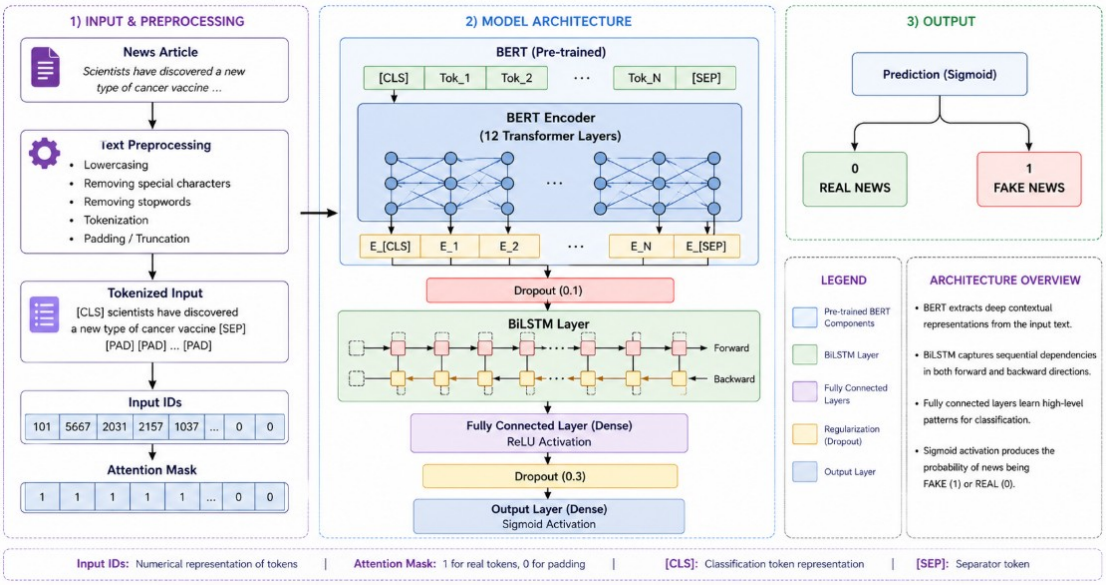


Figure 3-2: Detailed architecture of the proposed BERT + BiLSTM fake news detection model

As shown in the figure, the input news article is first tokenized and normalized (lowercasing, special-character removal, stopword removal, tokenization, padding/truncation), producing a fixed-length sequence of token IDs together with an attention mask. This tokenized sequence, prefixed with a [CLS] token and suf-

fixed with a [SEP] token, is passed through a pre-trained BERT encoder consisting of 12 Transformer layers, which produces a contextual embedding for every token position ($E_{[CLS]}, E_1, \ldots, E_N, E_{[SEP]}$).

Rather than discarding all but the [CLS] embedding, as is common in simple BERT fine-tuning setups, the proposed model passes the entire sequence of contextual embeddings through a dropout layer (rate 0.1) and then into a BiLSTM layer, which processes the sequence in both forward and backward directions to capture sequential dependencies over BERT's already-contextualized representations. The output of the BiLSTM layer is passed through a fully connected (dense) layer with a ReLU activation, followed by a second dropout layer (rate 0.3), and finally an output dense layer with a sigmoid activation that produces the probability of the input being fake (1) or real (0).

the two model families have complementary strengths that motivate the hybrid design explored here. BERT excels at capturing context and semantics within text through its bidirectional Transformer design and self-attention mechanism, but its primary mechanism for aggregating sequence-level information is attention rather than explicit sequence modelling of the kind performed by recurrent networks.

BiLSTM, on the other hand, is purpose-built for modelling sequences and capturing temporal/positional dependencies in text, processing its input in both forward and backward directions. In the proposed hybrid framework, BERT acts as a contextual feature extractor that produces informative token-level embeddings; rather than depending solely on the [CLS] token, the entire sequence of contextual embeddings generated by BERT is fed into the BiLSTM layer, allowing the network to layer explicit sequential modelling on top of BERT's already rich contextual representations.

Consequently, the resulting BERT + BiLSTM framework is designed to benefit from both contextual embeddings (BERT) and sequential learning (BiLSTM). The comparison uses the training configuration reported in Table 3.2.

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